Lab Report

Computer Architecture Laboratory

Assignment 5

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CS24BT060

1 Configuration

- Main Memory Latency: 40 cycles
- ALU Latency: 1 cycle
- Multiplier Latency: 4 cycles
- Divider Latency: 10 cycles

2 Results

Program	Number of Cycles Taken	Throughput (IPC)
even-odd.asm	251	0.0239
descending.asm	14945	0.0185
fibonacci.asm	3817	0.0204
palindrome.asm	2282	0.0215
prime.asm	1373	0.0211

3 Observations

When operation latencies are taken into account, the total number of execution cycles increases significantly compared to the previous assignment. In particular, the high main memory latency introduces long pipeline stalls during memory operations.

As a result, the overall IPC decreases since the processor spends more cycles waiting for operations to complete. The interlock mechanism correctly stalls the pipeline when dependent instructions require results that are not yet available, ensuring correct execution despite the increased latencies.

These results highlight the importance of optimizing memory accesses, which can be improved through techniques such as introducing a cache to reduce effective memory latency.